NAME:
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INSTRUCTOR:

# LINEAR ALGEBRA

## SPRING 2026

### QUIZ 7 (12th March 26)

**Max Marks: 10**

**Time: 10 minutes**

Q1. The following sets are **not** vector spaces. State which axiom(s) fail in each case and why.

(a) The set of all pairs of real numbers of the form $(x, y)$ , where $x \geq 0$, with the standard operations on $R^2$. [2]

(b) The set of all pairs of real numbers $(x, y)$ with the operations defined as: [4]

$$(x, y) + (x', y') = (x + x', y + y')$$
$$\text{and } k(x, y) = (3kx, 3ky)$$

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Q.2 Determine which of the following are subspaces of $\boldsymbol{M_{nn}}$. [2 each]

(a) All $n \times n$ matrices $A$ such that the linear system $A\boldsymbol{x} = \boldsymbol{0}$ has only the trivial solution.

(b) All $n \times n$ matrices $A$ such that $BA = AB$ for a fixed $n \times n$ matrix $B$.

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**Time: 10 minutes**

Q. Let the set $V$ consist of a single object, the zero vector, which is denoted by $\vec{\mathbf{0}}$. Define:

$$\vec{\mathbf{0}} + \vec{\mathbf{0}} = \vec{\mathbf{0}} \text{ and } k\,\vec{\mathbf{0}} = \vec{\mathbf{0}}$$

for all scalars $k$.

(a) Check if all the **vector space axioms** are satisfied under these operations.

(b) Using your answer to (a) above, what conclusion can be drawn about $V$ ?

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# QUIZ 7 SOLUTION

Q1. (a)

**Solution:**

Not a vector space as one axiom fails here: There exists an object $(x_k, y_k)$ such that $(x, y) + (x_k, y_k) = 0$
This can't be true as $x \in \mathbb{R}^+$ hence $x_k \in \mathbb{R}^-$ which does not exist in this vector space.

(b) Axioms 9 and 10 fail

(9) $k(l\underline{u}) = k[l(x,y)]$
$= k(2lx, 2ly) = (4lkx, 4kly)$ — ①
$(kl)\underline{u} = (kl)(x,y) = (2klx, 2kly)$ — ②
FROM ① AND ② $k(l\underline{u}) \neq (kl)\underline{u}$
SO NOT A VECTOR SPACE.

(X) $1\underline{u} = 1(x,y) = (2x, 2y) \neq (x,y)$
$\Rightarrow 1\underline{u} \neq \underline{u}$
$\because k(x,y) = (2kx, 2ky)$

Q.2 (a) It is a subspace

Closed under addition

$$BA_1 + BA_2 = A_1B + A_2B \Rightarrow B(A_1 + A_2) = (A_1 + A_2)B \Rightarrow BA_3$$

Closed under scalar multiplication.

$$B(kA_1) = (kA_1)B = k(AB)$$

(b)

**Solution:** It is not a subspace.
Since $Ax = 0$ has only trivial solution implies $A^{-1}$ exists.
Let $u = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $v = \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix}$ are both invert-able but $u + v$ is not invert-able.

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# QUIZ 8 SOLUTION

Let $V = \{\vec{0}\}$ be a set consisting of a single vector $\vec{0}$. The operations are defined as

$$\vec{0} + \vec{0} = \vec{0}, \qquad k\vec{0} = \vec{0}$$

for every scalar $k$.

We verify the vector space axioms.

1. **Closure under addition:** Since $\vec{0} + \vec{0} = \vec{0} \in V$, the set is closed under addition.
2. **Commutativity of addition:**
$$\vec{0} + \vec{0} = \vec{0} = \vec{0} + \vec{0}.$$
3. **Associativity of addition:**
$$(\vec{0} + \vec{0}) + \vec{0} = \vec{0} + \vec{0} = \vec{0}$$
and
$$\vec{0} + (\vec{0} + \vec{0}) = \vec{0} + \vec{0} = \vec{0}.$$
4. **Additive identity:** The vector $\vec{0}$ acts as the additive identity since
$$\vec{0} + \vec{0} = \vec{0}.$$
5. **Additive inverse:** The additive inverse of $\vec{0}$ is itself because
$$\vec{0} + \vec{0} = \vec{0}.$$
6. **Closure under scalar multiplication:** For every scalar $k$,
$$k\vec{0} = \vec{0} \in V.$$
7. **Distributive properties:** For scalars $a, b$,
$$(a + b)\vec{0} = \vec{0} = a\vec{0} + b\vec{0}.$$
For vectors,
$$a(\vec{0} + \vec{0}) = a\vec{0} = \vec{0} = a\vec{0} + a\vec{0}.$$
8. **Compatibility with scalar multiplication:**
$$a(b\vec{0}) = a\vec{0} = \vec{0} = (ab)\vec{0}.$$
9. **Identity scalar:**
$$1\vec{0} = \vec{0}.$$

Hence all vector space axioms are satisfied.

**Conclusion:**

Since $V$ satisfies all the vector space axioms and contains only the zero vector, $V$ is called the **zero vector space** (or **trivial vector space**).